

Aggregate Functions continued - 18/06/2026

↳ summarize data to extract specific information

↳ perform calculations on multiple rows of a specific column and returns a single value / single value per group

↳ consolidate the data

① COUNT () - we count unique identifiers (ID)

② SUM () - we sum numeric data (revenue)

③ AVG () - we average numeric data (revenue)

④ MIN () - looks for min value in a column

⑤ MAX () - looks for max value in a column

- GROUP BY statement gender, race, category

SELECT categorical column
Aggregate_Function (column) AS descriptive_column_name
FROM table_name
GROUP BY categorical_column; numerical column

↳ We group by the column that is not aggregated

- HAVING statement

↳ it is used to filter data that is aggregated

↳ we apply the HAVING statement to data that is already aggregated

↳ it is used to filter grouped results after using a GROUP BY

WHERE filters rows before grouping

HAVING filters groups after grouping

Sales

cust_id	cust_name	product	amount
101	Rothi	Laptop	5000
101	Rothi	Mouse	500
102	Abel	Keyboard	100
102	Abel	Monitor	2500
103	Ciera	Mouse	500

- these columns are aggregated

- these columns are categorical, because they give diff cust names

Write an SQL query to find customers who spend more than 5000

1. First calculate how much each customer spent

SELECT cust_name,
SUM (amount) AS total_spent

FROM sales
GROUP BY cust_name;

cust_name total_spent

Rothi 5500

Abel 2600

Ciera 500

- Secondly you must filter it to the question

```
SELECT cust_name, SUM (amount) AS total_spent
FROM sales
GROUP BY cust_name;
HAVING SUM (amount) > 8000
```



cust_name	total_spent
Rothi	8200
Abel	8000